INCH-POUND

MIL-P-211E
28 May 1991
SUPERSEDING
MIL-P-211D
9 May 1990
DOD-P-51463(EA)
1 June 1989

MILITARY SPECIFICATION

PHOSPHORUS, RED, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four classes of red phosphorus.

1.2 Classification. Red phosphorus shall be of the following classes as specified (see 6.2):

Class 1 - Fine granulation
Class 2 - Extra fine granulation
Class 3 - Dust suppressed with lubricating oil
Class 4 - Dust suppressed with aliphatic alcohol alkoxylate

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

NN-P-71 - Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)

MILITARY

MIL-B-82647 - Bag, Conductive Plastic, Heat-Sealable, Flexible

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to the Government

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 199 - Department of Transportation Hazardous Materials Regulations

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(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders for the above regulation should cite "49 CFR 171 to 199".)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

INTERNATIONAL CIVIL AVIATION ORGANIZATION

Technical Instructions for the Safe Transport of Dangerous Goods by Air

(Application for copies should be addressed to the International Civil Aviation Organization, 1000 Sherbrooke Street West, Suite 400, Montreal, Quebec, Canada H3A 2R2)

INTERNATIONAL MARITIME ORGANIZATION

International Maritime Dangerous Goods Code

(Application for copies should be addressed to the International Maritime Organization, 101-104 Piccadilly, London, WIV OAE, England.)

ASTM STANDARDS

D 92 - Flash and Fire Points by Cleveland Open Cup
D 97 - Pour Point of Petroleum Oils
D 130 - Detection of Copper Corrosion from Petroleum Products by the Copper Strip Tarnish Test
D 445 - Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity)
D 974 - Acid and Base Number by Color-Indicator Titration
D 1193 - Reagent Water
D 1500 - ASTM Color of Petroleum Products (ASTM Color Scale)
D 2270 - Calculating Viscosity Index from Kinematic Viscosity at 40 and 100°C
E 11 - Wire-Cloth Sieves for Testing Purposes

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Class 1 and class 2 chemical and physical characteristics. Class 1 and class 2 red phosphorus shall conform to the applicable chemical and physical characteristics of table I when tested as specified therein.

TABLE I. **Class 1 and class 2 chemical and physical characteristics**

Characteristic	Percent by weight				Test Paragraph
	Class 1		Class 2		
	Minimum	Maximum	Minimum	Maximum	
Red phosphorus	99.0		99.0		4.2.4.1
White phosphorus		0.02		0.02	4.2.4.2
Acidity (as H_3PO_4)		0.10		0.10	4.2.4.3
Moisture		0.20		0.20	4.2.4.4
Particle size:					4.2.4.5
Retained on a No. 40 (425 micrometer) sieve		0.1			
Retained on a No. 100 (150 micrometer) sieve				2.0	

3.2 Class 3 chemical and physical characteristics. Class 3 red phosphorus shall be red phosphorus which has been uniformly coated with lubricating oil (as a dust suppressant). The lubricating oil shall be a refined petroleum product that is clean, homogenous, and free from foreign matter and offensive odor. The lubricating oil shall conform to the chemical and physical characteristics of table II when tested as specified therein. No other materials shall be added to the red phosphorus. The contractor shall certify that the requirements for the lubricating oil have been met. Class 3 red phosphorus shall conform to the chemical and physical characteristics of table III when tested as specified therein.

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TABLE II. **Lubricating oil chemical and physical characteristics**

Characteristic	Requirement	ASTM test method
Viscosity, kinematic at 37.8°C	17 to 20	D 445
Flash point, minimum, °C	149	D 92
Pour point, maximum, °C	-34	D 97
ASTM Color, maximum	3	D 1500
Neutralization number, maximum	0.10	D 974
Copper corrosion, ASTM classification, maximum at 100°C	1	D 130
Viscosity index, minimum	30	D 2270

TABLE III. **Class 3 chemical and physical characteristics**

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Red phosphorus	97.0		4.2.4.6
White phosphorus		0.02	4.2.4.7
Acidity (as H_3PO_4)		0.10	4.2.4.3
Volatile matter		0.25	4.2.4.10
Lubricating oil (dust suppressant) content	0.8	1.3	4.2.4.9
Particle size: Retained on a No. 80 (180 micrometer) sieve		1.0	4.2.4.5

3.3 **Class 4 chemical and physical characteristics.** Class 4 red phosphorus shall conform to the chemical and physical characteristics of table IV when tested as specified therein. The hydrophilic dust suppressant shall be a straight chain, primary aliphatic alcohol alkoxylate added during manufacture prior to drying; the contractor shall certify that this requirement has been met.

3.4 **Material Safety Data Sheets.** Material Safety Data Sheets for red phosphorus shall be prepared and submitted by the contractor in accordance with FED-STD-313 (see 6.3).

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TABLE IV. **Class 4 chemical and physical characteristics**

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Red phosphorus	97.0		4.2.4.6
White phosphorus		0.02	4.2.4.7
Acidity (as H_3PO_4)		0.10	4.2.4.3
Moisture		0.20	4.2.4.8
Dust suppressant content*	0.80	1.5	4.2.4.9
Particle size: Retained on a No. 100 (150 micrometer) sieve		2.0	4.2.4.5

*Dust suppressant shall be a straight chain, primary, aliphatic alcohol alkoxylate (see 6.3)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling

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procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.9.) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Quality conformance inspection

4.2.1 Lotting. A lot shall consist of the red phosphorus of one class produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.5).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be as specified in the contract (see 6.2).

4.2.2.2 For test (see 6.6). Sampling shall be conducted in accordance with table V. A representative specimen of 453 ± 3 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE V. **Sampling for test**

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

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4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled shipping container ready for shipment. Sample shipping containers shall be examined for the following defects:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Bag exterior or drum interior or exterior contaminated with red phosphorus

4.2.3.2 For test (see 6.6). Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 450 g. The composited specimen shall be tested as specified in the applicable portions of 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests (see 6.6). Water in accordance with ASTM D 1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Red phosphorus (for classes 1 and 2).

(a) Magnesia mixture. Dissolve 110 g of magnesium chloride in a small amount of water. Add 280 g of ammonium chloride and 700 milliliters (mL) of concentrated ammonium hydroxide. Dilute to 2,000 mL with water and allow to stand overnight. Filter into a large bottle equipped with a glass stopper.

(b) Specimen solution. Transfer 1.2 g of the specimen to a small tared vial. Place the vial in the flask of a reflux digestion apparatus consisting of a 500-mL flask and 305 millimeter (mm) condenser with a ground glass joint. Add 100 mL of water to the flask and attach it to the reflux component of the apparatus. Add 50 mL of concentrated nitric acid through the delivery tube. Heat the flask gently and continue heating until the reaction is complete. Cool the contents of the flask. Add 3 mL of bromine, bring the solution to a gentle boil, and continue boiling for no less than 30 minutes. Transfer the solution to a 600-mL beaker. Wash the reflux apparatus thoroughly and add the washings to the transferred solutions. Boil the solution gently to remove the excess bromine. Cool the solution and filter. Wash the precipitate, if any, and transfer the filtrate and washings to a 500-mL volumetric flask. Cool and dilute to the mark with water, mixing thoroughly.

(c) Procedure. Transfer a 50-mL aliquot of the specimen solution prepared in (b) to a 400-mL beaker. Add a filtered solution of 15 g of ammonium nitrate in 200

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mL of water. Add dropwise, while stirring, 30 mL of cold magnesia mixture prepared in (a). While stirring, add concentrated ammonium hydroxide to the solution at the rate of 25 mL for each 100 mL of solution. Allow to settle for no less than 2 hours while digesting on a steam bath. Filter the precipitate and wash with 25-percent ammonium hydroxide solution. Transfer the precipitate and filter paper to a tared porcelain crucible and moisten with a few drops of saturated ammonium nitrate solution. Heat over a low flame until the filter paper is charred. Break up any lumps with a platinum rod. Place the crucible in a muffle furnace and heat to 1,100° ± 10°C. Cool in a desiccator and weigh the crucible and magnesium pyrophosphate. Calculate the percent by weight red phosphorus in the specimen as follows:

$$\text{Percent red phosphorus} = \frac{27.86\ (A - B)}{W} - C - 0.317D$$

where: A = Weight of the crucible and magnesium pyrophosphate, in grams,
B = Tare weight of the crucible, in grams,
C = Percent white phosphorus (4.2.4.2),
D = Percent acidity (4.2.4.3), and
W = Weight of the specimen, in grams, in the aliquot.

4.2.4.2 White phosphorus (for classes 1 and 2). Set up a Soxhlet or equivalent extraction apparatus. Pack a small extraction thimble with cotton and place in the extraction tube. Add 100 mL of benzene to the extraction flask, connect the extraction flask to the extraction apparatus, and extract with benzene for 15 minutes. Cool the apparatus, discard the benzene, remove the thimble from the extraction tube, and remove the cotton from the thimble. Weigh, to the nearest milligram (mg), 5 g of specimen into the extracted thimble and pack the thimble tightly with the extracted cotton. Place the thimble in the extraction tube, add 100 mL of benzene to the extraction flask, connect the extraction flask to the extraction apparatus, and extract the specimen with benzene for no less than 2 hours. Cool the apparatus and drain the benzene from the extraction tube into the extraction flask. Filter the benzene from the extraction flask through double-washed medium weight filter paper (Whatman No. 40 or equivalent). Thoroughly rinse the extraction flask with several small portions of benzene and add the rinsings to the benzene being filtered. (Reserve the extraction thimble and contents for preparation of a blank.) Collect the filtered benzene in a 250-mL glass-stoppered Erlenmeyer flask, add 10 mL of 25-percent copper nitrate solution, stopper, and shake for no less than 30 minutes on a shaking machine. Place the flask on a steam bath and evaporate the benzene. Add 20 mL of bromine saturated concentrated nitric acid. Heat on a hot plate until all nitric acid and bromine fumes are expelled. Cool and filter through filter paper into a second 250-mL Erlenmeyer flask. Wash the first 250-mL flask with several small portions of hot water and add each portion to the solution being filtered. (The volume at this point should be approximately 100 mL.) Make the solution slightly alkaline (blue) with ammonium

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hydroxide and cool to room temperature. Make the solution slightly acid (green) with nitric acid. Add 25 mL of ammonium molybdate solution (0.75 g per 100 mL, in water). Agitate in a shaking machine for no less than 30 minutes to complete the precipitation. Filter the solution and discard the filtrate. Wash the precipitate with 3-percent sodium nitrate (or potassium nitrate) solution until neutral as determined by using pH paper. Transfer the filter paper and precipitate to a 250-mL glass-stoppered Erlenmeyer flask; add 50 mL of water. Agitate the flask, add 5 drops of phenolphthalein indicator solution, and titrate with 0.1*N* sodium hydroxide solution to the pink endpoint. Add 5 to 10 mL of the 0.1*N* sodium hydroxide solution in excess. Stopper the flask, shake well, and titrate the excess sodium hydroxide with 0.05*N* nitric acid solution. For the blank, re-extract the extraction thimble and contents with a fresh 100 mL of benzene for 2 hours. Treat the benzene extract as above. Calculate the percent white phosphorus as follows:

$$\text{Percent white phosphorus} = \frac{0.13\ [(AC - DF) - (BC - EF)]}{W}$$

where: A = Milliliters of sodium hydroxide solution used for specimen,
B = Milliliters of sodium hydroxide solution used for blank,
C = Normality of the sodium hydroxide solution,
D = Milliliters of nitric acid solution used for specimen,
E = Milliliters of nitric acid solution used for blank,
F = Normality of the nitric acid solution, and
W = Weight of specimen, in grams.

4.2.4.3 Acidity (as orthophosphoric acid) (for classes 1, 2, 3, and 4). Mix 10.0 g of specimen, weighed to the nearest 1 mg, with 5 mL of neutral methanol (or ethanol). Add three 25-mL portions of water to the mixture, decanting after each addition. Filter the decanted water solution through unwashed, retentive filter paper (Whatman No. 5 or equal). Add 3 drops of methyl orange and titrate with 0.1*N* sodium hydroxide solution to an endpoint of the same color as 3 drops of methyl orange in 100 mL of 0.5*M* potassium phosphate, monobasic. Run a blank parallel to the specimen. (If the blank, after adding the methyl orange, is yellow, the blank titer is zero; if the blank is pink, titrate to yellow.) Calculate the percent acidity as orthophosphoric acid as follows:

$$\text{Percent acidity} = \frac{9.8\ N\ (A - B)}{W}$$

where: N = Normality of the sodium hydroxide solution,
A = Milliliters of sodium hydroxide solution used for the specimen,
B = Milliliters of sodium hydroxide solution used for the blank, and
W = Weight of specimen, in grams.

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4.2.4.4 Moisture (for classes 1 and 2). Transfer 5 g of the specimen weighed to the nearest 0.1 mg to a watch glass or wideform drying bottle. Place over concentrated sulfuric acid in a desiccator and dry at room temperature to a constant weight. Calculate the percent by weight moisture as follows:

$$\text{Percent moisture} = \frac{100(A - B)}{W}$$

where: A = Weight of the specimen and container before drying, in grams,
B = Weight of the specimen and container after drying, in grams, and
W = Weight of the specimen, in grams.

4.2.4.5 Particle size (for classes 1, 2, 3, and 4). Transfer 100 g of the specimen to a tared sieve of the appropriate size (see tables I, III, and IV) and conforming to ASTM E 11. If the material is of a waxy consistency, moisten it with a few milliliters of ethanol or methanol. Wash the specimen through the screen with water until the quantity of material passing through the sieve does not exceed a trace. Should the waxy condition recur, add a few more milliliters of alcohol to the specimen. Dry the sieve over concentrated sulfuric acid. Calculate the percent of specimen retained as follows:

$$\text{Percent retained} = \frac{100(A - B)}{W}$$

where: A = Weight of the sieve and contents, after drying, in grams,
B = Tare weight of the sieve, in grams, and
W = Weight of specimen, in grams.

4.2.4.6 Red phosphorus (for classes 3 and 4).

(a) **TS 4.5 indicator (pH 4.0 to 4.6, orange to blue).** Dissolve 0.15 g of methyl red indicator and 1.40 g of bromocresol green indicator in 1 liter of ethanol. When solution is complete, neutralize with sodium hydroxide.

(b) **Extraction of dust suppressant.** Mix 30 g of specimen with 500 mL of methylene chloride in a 1-liter beaker for 30 minutes using a mechanical stirrer, then filter through a Whatman No. 1 filter paper or equal in a Buchner funnel under suction. Transfer the specimen to the funnel and wash with a total of 300 mL of methylene chloride. Suck the residue dry, transfer it to a large watch glass, and place it in a vacuum oven at 60°C for 10 minutes. Transfer the stripped specimen to a small screw-capped bottle.

(c) **Matter insoluble in aqua regia.** Weigh 10.0 g of stripped specimen to the nearest 0.1 mg into an 800-mL beaker. Transfer the beaker to a fume cupboard and add successively 100 mL of distilled water, 50 mL of nitric acid, and 50 mL of

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hydrochloric acid. Warm the specimen gently on a hot plate to initiate the reaction. When the reaction starts, remove the beaker from the hot plate, cover with a watch glass, and allow the reaction to proceed. When the reaction has subsided, rinse down the inside of the beaker with a small amount of distilled water. Evaporate to about 60 mL then add 50 mL of nitric and rinse down the inside of the beaker with a small amount of water, if necessary. Evaporate to about 60 mL. Dilute the mixture to about 600 mL with distilled water and heat to about 60°C. Filter the solution through a tared, fine sintered glass crucible, wash the precipitate well with hot water, and dry the crucible at 105°C for 1 hour. Cool the crucible in a desiccator and reweigh. Calculate the percent by weight matter insoluble in aqua regia as follows:

$$\text{Percent matter insoluble in aqua regia} = \frac{100A}{W}$$

where: A = Weight of residue in, grams, and
W = Weight of stripped specimen, in grams.

(d) Procedure. Transfer 5.0 g of stripped specimen weighed to the nearest 0.1 mg into a 400-mL beaker. Add 100 mL of 1*M* hydrochloric acid and heat the mixture to boiling. Filter the boiling solution through a tared, fine porosity, sintered glass crucible. Transfer all of the insoluble material into the crucible and wash with hot water until the filtrate is acid free to the TS 4.5 indicator prepared in (a) as indicated by a blue color. Wash the residue twice with 20-mL portions of methanol and dry the crucible at 60°C in a vacuum oven for 4 hours. Cool in a desiccator and reweigh. Calculate the percent by weight of red phosphorus as follows:

$$\text{Percent red phosphorus} = \left(\frac{100A}{W} - B\right)\left(\frac{100 - C}{100}\right)$$

where: A = Weight of residue in, grams,
B = Percent by weight matter insoluble in aqua regia determined in (c),
C = Percent by weight dust suppressant determined in 4.2.4.9, and
W = Weight of stripped specimen, in grams.

4.2.4.7 White phosphorus (for classes 3 and 4).

(a) Extraction of dust suppressant. Extract dust suppressant as specified in 4.2.4.6(b).

(b) Preparation of copper sulfate test paper. Soak pieces of filter paper in 10-percent weight per volume cupric sulfate pentahydrate solution and allow to air dry. Cut the dried paper into strips about 0.6 cm wide and 10 cm long.

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(c) **White phosphorus stock standard solution, 5,000 micrograms per milliliter.** Accurately weigh 1.000 g of white phosphorus into a 200-mL beaker. Add 100 mL of carbon disulfide and stir to dissolve. Transfer to a 200-mL volumetric flask and dilute to the volume mark with carbon disulfide. Store this solution in the dark and renew after 6 months.

(d) **White phosphorus standard solution, 500 micrograms per milliliter.** Pipet 20 mL of the 5,000 micrograms per milliliter (µg/mL) solution into a 200-mL volumetric flask and dilute to the volume mark with carbon disulfide. Store this solution in the dark and renew after 6 months.

(e) **White phosphorus comparison solutions.** Prepare 5, 10, 20, 30, and 40 µg/mL solutions of white phosphorus in carbon disulfide by transferring 1.0, 2.0, 4.0, 6.0, and 8.0 mL quantities, respectively, of the 500 µg/mL white phosphorus standard solution into separate 100-mL volumetric flasks and diluting to the volume marks with carbon disulfide. Other comparison solutions may be prepared as desired. Store these solutions in the dark and renew after 3 months.

(f) **Procedure.** Weigh 20 g of the stripped specimen into a 100-mL beaker and add approximately 50 mL of carbon disulfide. Shake and mix thoroughly. Transfer to a 100-mL volumetric flask and dilute to the volume mark with carbon disulfide. Determine the white phosphorus concentration by immersing strips of the copper sulfate test paper prepared in (b) in the specimen solution and in the comparison solutions prepared in (e) for 1 minute. Visually compare the brown stain produced by the specimen solution to those produced by the comparison solutions. Calculate the percent by weight white phosphorus as follows:

$$\text{Percent white phosphorus} = \frac{100AB}{W}$$

where: A = Amount of white phosphorus found in specimen solution, in micrograms per milliliter,
B = Volume of specimen, in milliliters, and
W = Weight of specimen, in grams.

(g) **Disposal of carbon disulfide and red phosphorus.** Decant the carbon disulfide into an evaporating dish. Incline the volumetric flask and allow the carbon disulfide to evaporate completely from the dish and the volumetric flask. Rinse the red phosphorus out of the volumetric flask with water into the red phosphorus residues container.

4.2.4.8 Moisture (for class 4). Tare a clean, dry watch glass and record all weights to the nearest 0.1 mg. Add 5 g of specimen to the watch glass and record the gross weight. Place the watch glass and specimen in a drying oven at 103°C for 2 hours. At the end

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of the drying time, place the watch glass and specimen in a desiccator. Cool and weigh to a constant weight. Calculate the percent by weight moisture as follows:

$$\text{Percent moisture} = \frac{100(A - B)}{W}$$

where: A = Weight of specimen and container before drying, in grams,
B = Weight of specimen and container after drying, in grams, and
W = Weight of specimen, in grams.

4.2.4.9 Dust suppressant (for classes 3 and 4). Weigh 5.0 g of specimen into a 150-mL beaker and add 50 mL of methylene chloride and a 1-inch stirring bar. Cover with a watch glass and stir for 30 minutes. Filter the solution through a Whatman No. 42 or equal filter paper into a clean, tared, glass evaporating dish. Wash the beaker and paper three times into the evaporating dish using a total of 50 mL of methylene chloride. Place the evaporating dish containing the filtrate and washings on a steam bath and evaporate to about 3 mL of solution. Do not heat the residue to dryness as loss of dust suppressant may occur. Remove from the steam bath and evaporate the remaining methylene chloride in an air stream. Dry off traces of outside moisture, cool in a desiccator, and weigh. Calculate the percent by weight dust suppressant as follows:

$$\text{Percent dust suppressant} = \frac{100(A - B)}{W}$$

where: A = Weight of dish and residue, in grams,
B = Weight of dish, in grams, and
W = Weight of specimen, in grams.

4.2.4.10 Volatile matter (for class 3). Heat a flat-bottomed glass or aluminum dish with a loosely fitting cover in a hot water bath for at least 30 minutes. Cool the dish and cover in a desiccator for 30 minutes and weigh them together. Immediately transfer 5 g of specimen to the dish, cover it, and accurately weigh. Place the dish and cover, with cover removed, in the vacuum desiccator over freshly activated silica gel and maintain a maximum 10 millimeters of mercury absolute pressure for 18 to 24 hours. Release the vacuum slowly and pass the incoming air through a suitable drying vessel filled with activated silica gel. Replace the dish cover and weigh. All weighings should be accurate to 0.1 mg. Calculate the percent by weight volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100(A - B)}{(A - C)}$$

where: A = Weight of dish, cover, and specimen before evaporation, in grams,

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contain a minimum of 250 pounds net

B = Weight of dish, cover, and specimen after evaporation, in grams, and
C = Weight of empty dish and cover, in grams.

5. PACKAGING

5.1 Packaging. In addition to the packaging requirements in this specification, red phosphorus shall be packaged in accordance with the applicable packaging, marking, and labeling requirements of the latest editions of the International Civil Aviation Organization Technical Instructions for Safe Transportation of Dangerous Goods by Air (ICAO-TDGA), the International Maritime Organization International Maritime Dangerous Goods Code (IMO-IMDGC), and 49 CFR 171 to 199. To the extent that there is conflict between the packaging requirements in this specification and the requirements of ICAO-TDGA and IMO-IMDGC, the provisions in ICAO-TDGA and IMO-IMDGC shall take precedence. All performance test requirements shall be supported by certificates and reports attesting to the date and the data results obtained from performance oriented packaging (POP) testing. The contractor, as the certifier, shall be responsible for providing testing services. Contractors who do not self certify shall be responsible for assuring that third party sources providing performance testing are in fact registered with the Department of Transportation. All certificates and reports shall be available for inspection by an authorized Government representative for a period of no less that 3 years.

5.2 Packing. Packing shall be level A, B, C, or for interplant shipment as specified (see 6.2).

5.2.1 Level A. A quantity of 100 (+0.5 or -0) pounds (lb) of red phosphorus shall be packed in a new, removable-head, 11-gallon steel drum conforming to Department of Transportation (DOT) Specification 6B or 37A, provided with a polyethylene bag liner. The inner and outer surfaces of the drum shall be prepared for coating and painting in accordance with PPP-D-729. All inner surfaces shall be covered with two coats of high-baked epoxy-phenolic resin totalling no less than 0.025 mm. All outer surfaces of the drum shall be painted as specified in PPP-D-729. The bag liner shall conform to MIL-B-82647 and shall be formed from film having a minimum thickness of 0.003 inch. The bag liner shall be closed by tying or knotting. There shall be no evidence of red phosphorus or moisture on the interior or exterior of the drum. The drum shall be closed in accordance with DOT regulations.

5.2.2 Level B. A quantity of 100 (+0.5 or -0) lb of red phosphorus shall be packed in a removable-head, gasketed-head steel drum which provides moisture resistance and is furnished with an electrically conductive close-fitting bag liner. The drum shall conform to DOT Specification 6B. The bag shall conform to MIL-B-82647 and shall be formed from film having a minimum thickness of 0.003 inch. The bag liner shall be closed by tying or knotting. There shall be no evidence of red phosphorus on the exterior of the bag

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or the interior or exterior of the drum. The drum shall be closed in accordance with DOT regulations.

5.2.3 **Level C.** A quantity of red phosphorus as specified (see 6.2) shall be packed in accordance with DOT regulations in a steel drum fitted with a polyethylene bag liner. The bag liner shall conform to MIL-B-82647 and shall be formed from film having a minimum thickness of 0.003 inch. The bag liner shall be closed by tying or knotting. The drum shall conform to DOT Specification 6B or 37A.

5.2.4 **Interplant shipment (see 6.7)** Military packaging is not applicable to interplant shipments. The red phosphorus shall be packed to provide adequate protection from physical damage.

5.3 **Unitization.** When specified (see 6.2), shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147. Pallets shall conform to type IV of NN-P-71. A horizontal strap shall be applied to each tier and tie-down straps parallel to the 1-meter dimension only of the pallet shall be used. When specified (see 6.2), the pallet and any other wood storage aids shall be treated with wood preservative as specified in NN-P-71.

5.4 **Marking.**

5.4.1 **Levels A, B, and C shipments.** Shipping containers shall be marked in accordance with DOT regulations and MIL-STD-129. The DOT "Flammable Solid" label shall be used on all containers. Each drum shall bear the special prominent marking: "When emptying, clamp the bag to the bare inner surface of the drum and make good electrical connection to ground."

5.4.2 **Interplant shipment.** Shipping containers shall be marked in accordance with DOT regulations. The DOT "Flammable Solid" label shall be used on all containers. Containers shall be marked to identify the product in addition to providing any other required data during shipment from the source to the first receiving activity.

6. **NOTES**

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** Red phosphorus is intended for use in pyrotechnic mixtures. Use of classes 3 and 4 red phosphorus will reduce dust during pyrotechnic processing.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
(c) Class of red phosphorus required (see 1.2)
(d) Sampling for examination of packaging (see 4.2.2.1)
(e) Level of packing required (see 5.2)
(f) Unit quantity required for level C packing (see 5.2.3)
(g) Unitization and preservative, if required (see 5.3)

6.3 Aliphatic alcohol alkoxylate dust suppressant. Alkasurf LA-EP 25LF, manufactured by Alkaril Chemicals LTD, has been found to meet the material requirements.

6.4 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent mailing addresses for submission of data are listed in FED-STD-313.

6.5 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.6 Sampling and testing precaution. This specification may involve hazardous materials, operations, and equipment. This specification does not purport to address all of the safety problems associated with its use. It is the responsibility of the user of this specification to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

6.7 Interplant shipment. Packing and marking for interplant shipment is used for supplies and materials that do not directly enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or to a military arsenal, plant, or other activity for evaluation, immediate use, or further processing as specified in the applicable contract.

6.8 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.9 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

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6.10 Supersession data. DOD-P-51463(EA), dated 12 February 1980, with Amendment 1, dated 9 June 1982, and Notice 1, dated 1 June 1989, is superseded by the class 3 red phosphorus of this specification. Future acquisition of the the oiled red phosphorus formerly acquired under DOD-P-51463(EA) should made by specifying class 3 of MIL-P-211.

6.11 Changes from previous issues. The margins of this specification are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.12 Subject term (key word) listing.

Acidity
Dust suppressant
Moisture
Particle size
Volatile matter
White phosphorus

Custodians:

Army - EA
Navy - OS

Review activities:

Army - AR, MD
Navy - AS
DLA - GS

Preparing activity:

Army - EA

Project No. 6810-1244

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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DOCUMENT TITLE

NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

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a. NAME (*Last, First, Middle Initial*)		b. ORGANIZATION	
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NAME	b. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON/DSN
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